

The DRanking Protocol: Verifiable Distributed Ranking and Sybil-Bounded Admission for Structured Peer-to-Peer Networks

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Abstract

DRanking converts locally observed peer behavior in an open structured peer-to-peer network into a globally agreed, verifiable service weight that gates admission to protocol roles. Identity creation is free, so no mechanism bounds adversarial count [1]; the protocol instead bounds adversarial weight and prices it. The instruments are: counterparty-signed service receipts in place of unsigned counters; a seed-anchored trust flow $t_{k+1} = \alpha C_k^T t_k + (1 - \alpha)s$ obeying conservation, attenuation, and revocation laws, with a proved bound $T_k(A) \leq \alpha g / (1 - \alpha)$ on Sybil mass; trust-weighted verifiable random sortition in place of oracle sampling; a proof-of-authority ledger trusted for ordering only, with equivocation proofs as first-class slash objects; and a circuit-friendly epoch transition folded by Nova-family incrementally verifiable computation, giving constant-cost verification to browser nodes. Rank is soulbound and slashable; reward is a separate transferable asset; the authority-to-stake transition is conditioned on a security-budget crossover.

I. Introduction

The 2023 draft of this protocol scored behavior per identity. Identities are free; every component secured against lying nodes fell to numerous nodes. This rewrite keeps the draft's decomposition — local evidence, global aggregation, sampling, ledger, incentives — and replaces the quantity being secured:

$$\begin{aligned} \text{cost}(\text{did}) &= 0, \\ |\text{Adv} \cap V| &\leq \infty, \\ \text{secure}(| \cdot |) &= \perp \quad [1], \\ \text{secure} &:= \text{secure}(T(\cdot)), \\ T(A) &= \sum_{v \in A} t(v), \\ \text{claim} &= \text{price}(T(\text{Adv}) \geq \tfrac{1}{3}). \end{aligned}$$

Sections II–IX construct t , its evidence, its sampling, its ledger, and its economy; Section X tabulates the attack surface. Each mechanism is a term of price.

II. Model

Definition 1 (Substrate).

$$\begin{aligned} R &= \mathbb{Z}/2^{160}\mathbb{Z} \quad [2], [3], \\ \text{did} &: K \rightarrow R, \quad \text{self-certifying}, \\ \text{link} &\subseteq V \times V, \quad \text{authenticated}, \\ \text{msg} &= \langle m, \sigma_{\text{origin}} \rangle, \\ \text{cost}(\text{did}) &= 0, \\ \text{cost}(\text{link}) &= \varepsilon > 0. \end{aligned}$$

Definition 2 (Epoch).

$$k \in \mathbb{N}, \quad \text{epoch}(k) = [\text{fin}(\Sigma_k), \text{fin}(\Sigma_{k+1})),$$

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delimited by ledger finality (Definition 11).

Assumption 1 (Adversary).

$$\begin{aligned} |\text{Adv}| &\leq \infty, \\ \text{collude, grind, patience} &\in \text{Cap}(\text{Adv}), \\ \text{byzantine} &\in \text{Cap}(\text{Adv}), \\ \text{forge}(\sigma_h), h \notin \text{Adv} &\notin \text{Cap}(\text{Adv}), \\ \text{break}(\text{VRF}) &\notin \text{Cap}(\text{Adv}). \end{aligned}$$

Assumption 2 (Honest weight).

$$T_k(\text{Adv}) < \tfrac{1}{3} T_k(V) \quad \forall k.$$

No bound is assumed on $|\text{Adv}|$.

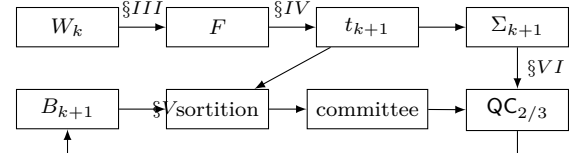


Figure 1. Epoch cycle: receipts fold into trust, trust seats the committee, the committee finalizes the ledger and threshold-signs the beacon.

III. Service Receipts

Unsigned counters are first-person testimony; aggregated, they are hearsay. The unit of evidence is a countersigned receipt.

Definition 3 (Receipt).

$$\begin{aligned} \rho &= \langle \text{srv}, a, b, k, n, \sigma_b \rangle, \\ \text{srv} &\in \{\text{relay, store, probe}\}, \\ a &= \text{server}, \quad b = \text{signer/served}, \\ \text{valid}(\rho) &\equiv \text{Vrfy}(k_b, \sigma_b, \text{srv} \| a \| b \| k \| n) = \top \\ &\quad \wedge (b, k, n) \text{ fresh}. \end{aligned}$$

Proposition 1 (Forgery surface).

$$\text{valid}(\rho) \wedge b \notin \text{Adv} \Rightarrow \rho \text{ issued by } b.$$

Proof. Immediate from $\text{forge}(\sigma_b) \notin \text{Cap}(\text{Adv})$ (Assumption 1). Manufacturable receipts are exactly those with $b \in \text{Adv}$ (self-dealing); these are neutralized by signer-trust weighting (Section IV). $\square$

Definition 4 (Receipt graph).

$$\begin{aligned} \text{val}(\rho) &= \mu_{\text{srv}(\rho)} \cdot \text{units}(\rho), \quad \mu \text{ published per kind,} \\ W_k &\in \mathbb{R}_{\geq 0}^{V \times V}, \\ W_k[b, a] &= \sum \{\text{val}(\rho) : \rho \text{ valid, sig } b, \text{ srv } a, \text{ ep } k\}, \\ W_k[b, \cdot] \mathbf{1} &\leq w_{\max}. \end{aligned}$$

The cap makes countersigning a budgeted act; the protocol certifies performed service, never synthetic load (cf. Constraint 1).

IV. Trust Algebra

Definition 5 (Seed vector).

$$\begin{aligned} s &\in [0, 1]^V, \quad \sum_v s_v = 1, \\ \text{supp}(s) &= S \text{ (authority set)}. \end{aligned}$$

Law 1 (Conservation).

$$C_k[b, a] = \frac{W_k[b, a]}{\max(w_{\max}, W_k[b, \cdot] \mathbf{1})}, \quad C_k[b, \cdot] \mathbf{1} \leq 1.$$

Law 2 (Attenuation).

$$\alpha \in (0, 1), \quad \text{flow}(\ell \text{ hops}) \leq \alpha^\ell.$$

Law 3 (Revocation).

$$\begin{aligned} \text{slash}(v) &\Rightarrow t(v) := 0, \quad v \notin \text{carrier}(t), \\ t_{k+1} &= f(W_k, t_k) \text{ (re-derived per epoch),} \\ s &\text{ withdrawable per seed.} \end{aligned}$$

Definition 6 (Trust update).

$$t_{k+1} = \alpha C_k^\top t_k + (1 - \alpha) s. \quad (1)$$

One power-iteration step per epoch toward $t^* = (1 - \alpha)(I - \alpha C^\top)^{-1} s$ [4], [5]; the fixpoint is approached along chain time, never computed within an epoch (cf. Definition 13).

Invariant 1 (Mass).

$$\sum_v t_0(v) = 1 \Rightarrow \sum_v t_k(v) \leq 1 \quad \forall k.$$

Proof.

$$\begin{aligned} \sum_v t_{k+1}(v) &= \alpha \mathbf{1}^\top C_k^\top t_k + (1 - \alpha) \\ &\leq \alpha \mathbf{1}^\top t_k + (1 - \alpha) \quad (\text{Law 1}) \\ &\leq \alpha + (1 - \alpha) = 1 \quad (\text{induction}). \end{aligned}$$

$\square$

Proposition 2 (Sybil influence bound). Partition $V = H \uplus A$, $S \subseteq H$. Let

$$g_k = \sum_{b \in H, a \in A} C_k[b, a] t_k(b), \quad g = \sup_k g_k.$$

Then

$$T_k(A) = \sum_{v \in A} t_k(v) \leq \frac{\alpha g}{1 - \alpha} \quad \forall k.$$

Proof. $S \cap A = \emptyset$ kills the seed term over A . Splitting (1) by origin region and applying Law 1 to internal circulation,

$$T_{k+1}(A) \leq \alpha(g_k + T_k(A)).$$

From $T_0(A) = 0$,

$$T_k(A) \leq \alpha g \sum_{i \geq 0} \alpha^i = \frac{\alpha g}{1 - \alpha}.$$

$\square$

$T(A)$ is linear in honest endorsement granted (g), independent of $|A|$. Defense concentrates on g : endorsement spends budget (Law 1), and detection claws it back (Law 3).

Proposition 3 (Farming price). Instantiating Proposition 2 at $A = \{v\}$: sustaining weight w requires per-epoch honest endorsement

$$g_v \geq \frac{1 - \alpha}{\alpha} w \quad \text{for every epoch the weight is held,}$$

so

$$\text{farm}(w, \ell \text{ epochs}) \geq \ell \cdot \text{cost}\left(\frac{1 - \alpha}{\alpha} w\right) + \text{Pr}[\text{detect}] \cdot w,$$

the second term by Law 3: one accepted equivocation forfeits the stock. This is the **farm** function of Proposition 13, made explicit.

Proof. $T_k(\{v\}) \leq \alpha g_v / (1 - \alpha)$; rearrange, multiply by holding duration, add the slash expectation. $\square$

Proposition 4 (Geometric convergence). For stationary $C_k = C$, the map $\Phi(x) = \alpha C^\top x + (1 - \alpha)s$ satisfies

$$\|\Phi(x) - \Phi(y)\|_1 \leq \alpha \|x - y\|_1,$$

hence t^* is the unique fixpoint and

$$\|t_k - t^*\|_1 \leq \alpha^k \|t_0 - t^*\|_1.$$

Proof.

$$\begin{aligned} \|C^\top(x - y)\|_1 &= \sum_a |\sum_b C[b, a](x_b - y_b)| \\ &\leq \sum_b (\sum_a C[b, a]) |x_b - y_b| \leq \|x - y\|_1 \end{aligned}$$

by Law 1; Banach on the affine contraction. Under drifting C_k the iterate tracks the moving fixpoint with lag $O(\alpha^j)$ per j epochs of stability. $\square$

Definition 7 (Soulbound rank).

$$\begin{aligned} \text{rank}_k(v) &= t_k(v), \\ \text{transfer} &\notin \text{Ops}(\text{rank}), \\ \text{rank} &\xrightarrow{\text{slash}} 0, \\ \text{rank} &\xrightarrow{\text{epoch}} \text{re-derived (1)}. \end{aligned}$$

(Cf. Proposition 13.)

Definition 8 (Dilution schedule).

$$\begin{aligned} \text{vol}_k &= \sum_{i \leq k} \mathbf{1}^\top W_i \mathbf{1}, \\ 1 - \alpha_k &= D(\text{vol}_k), \quad D \text{ non-increasing, published,} \\ \bar{\alpha} &= \sup_k \alpha_k < 1, \\ t_0 &= s \text{ (cold start).} \end{aligned}$$

Propositions 2–4 hold under the time-varying schedule with $\alpha := \bar{\alpha}$.

Definition 9 (Threshold authority).

$$\begin{aligned} \text{Ops}(S) &= \{\text{withdraw}, \text{rotate}, \text{endorse}_0\}, \\ o \in \text{Ops}(S) &\Rightarrow \text{QC}_\theta(S)(o), \quad \theta \geq \frac{2}{3}, \\ |\text{compromised}| < \theta|S| &\Rightarrow \text{Cap}(\text{Adv}) \cap \text{Ops}(S) = \emptyset. \end{aligned}$$

V. Sortition

Proposition 5 (Statistical undetectability). Model $\text{did} = \text{h}(\text{pk})$, h a random oracle. For the adversarial placement strategy “sample fresh keys,” honest and adversarial positions are identically distributed on R :

$$\Delta_{\text{TV}}(\text{pos}(H \oplus A), \text{pos}(H')) = 0 \quad \forall |A|,$$

hence every position-based test (entropy, Kolmogorov–Smirnov) has distinguishing advantage 0 against Sybil density.

Proof. $\text{h}(\cdot)$ uniform on R for every key, honest or adversarial; a union of i.i.d. uniforms is distribution-invariant in its partition. $\square$

Placement tests detect biased sampling; the threat is cheap identity. Validation must be cryptographic and economic, and a global random seed is a bootstrap circularity (a randomness beacon presupposes the committee being selected). Both are removed by sortition [6], [7].

Definition 10 (Beacon).

$$\begin{aligned} \text{TSig}_S &: \theta\text{-of-}|S|, \text{ unique [8]}, \\ |\{\sigma : \text{TVrfy}(k, \sigma) = \top\}| &= 1 \quad \forall k, \\ B_k &= \text{h}(\text{TSig}_S(k)). \end{aligned}$$

Proposition 6 (Beacon unbiasedness).

$$B_k = \text{fn}(\text{gpk}, k) \wedge \text{predict}(B_k) \Rightarrow \theta \text{ shares},$$

i.e. no participant choice influences B_k . A header-hash beacon $\text{h}(\text{fin}(\Sigma_k))$ is excluded: proposers grind B_k through content selection.

Proof. Uniqueness leaves a single valid σ per k : the signing protocol has no free variable to grind. Unpredictability below θ shares is the threshold-signature security game. $\square$

Algorithm 1 Trust-weighted sortition (node v , epoch k , role r)

- 1: $B_k \leftarrow \text{h}(\text{TSig}_S(k))$ $\triangleright$ Definition 10
 - 2: $(y, \pi) \leftarrow \text{VRF}_{\text{sk}_v}(B_k \| r)$
 - 3: if $y/2^{|y|} < \tau \cdot t_k(v)$ then
 - 4: v holds r in epoch k ; publishes π
-

Proposition 7 (Grinding closure).

$$\begin{aligned} \text{sk}_v \text{ fixed} &\prec B_k \text{ exists,} \\ B_k \text{ unbiased} &\quad (\text{Proposition 6}), \quad \Rightarrow \text{grind} = \perp. \end{aligned}$$

Proof. Both VRF inputs are committed before either party can condition on the output; unpredictability from Assumption 1 and Proposition 6. $\square$

Proposition 8 (Committee safety). Let seats be independent indicators with $\mathbb{E}[X] = \tau$, $\mathbb{E}[X_A] = \tau T(A)$, $T(A) < \frac{1}{3}$. By Chernoff–Hoeffding,

$$\Pr[X_A \geq \frac{\tau}{3}] \leq \exp\left(-\tau D\left(\frac{1}{3} \parallel T(A)\right)\right),$$

with D the binary relative entropy: committee dishonesty decays exponentially in committee size. Sortition is the transport of Assumption 2 from weight to seats; no reported quantity exists for a coalition to steer, so the draft’s median game is vacuous by construction.

VI. Admission Ledger

Definition 11 (Ledger).

$$\begin{aligned} \Sigma_k &= \langle \text{rt}(t_k), \text{rt}(V_k), \text{rt}(\text{sl}_k), \text{rt}(\text{bal}_k), \text{ver} \rangle, \\ \text{fin}(\Sigma_k) &\equiv \text{QC}_{2/3}(S_k) \quad [9], [10], \\ \text{content} &= \{\text{rt}(W_k)\} \cup \text{sl} \cup \Delta V \cup \text{issue}, \\ \text{data}(W_k) &\in \text{Overlay storage (not on chain)}. \end{aligned}$$

Obligation 1 (Minimal trust).

$$\begin{aligned} \forall o \in \text{content} : \text{Verify}(o) &\perp \text{ honest}(S), \\ \text{Cap}(\text{quorum}) &= \{\text{order}, \text{censor}\}, \\ \text{forgeable}(\text{quorum}) \ni o &\Rightarrow \text{design defect}. \end{aligned}$$

Definition 12 (Equivocation proof).

$$\begin{aligned} e &= \langle m_1, \sigma_v(m_1), m_2, \sigma_v(m_2) \rangle, \\ \text{slot}(m_1) = \text{slot}(m_2) \wedge m_1 \neq m_2 &\Rightarrow \text{slash}(v), \end{aligned}$$

verification assumption-free; e is a first-class ledger object, submittable by anyone.

Obligation 2 (Force inclusion).

$$\text{ack}_S(x) \wedge x \notin \Sigma_{k+\Delta} \Rightarrow \text{ack}_S(x) \in \text{sl (quorum slash)},$$

bounding censorship, the residual validator power, by latency Δ .

Equivocation defeats patience: months of farmed standing forfeit on one contradiction, so $\mathbb{E}[\text{farm}] \downarrow$ with detection probability.

VII. Folded State

Browser wasm nodes can neither replay history nor audit a validator signature; the chain must be succinct [11].

Definition 13 (Epoch transition).

$$F(\Sigma_k, \text{rt}(W_k)) = \Sigma_{k+1},$$

Algorithm 2 F — pure, deterministic, per epoch

- 1: $\text{sl}_{k+1} \leftarrow \text{sl}_k \cup \{\text{verify}(e) : e \text{ submitted}\}$
 - 2: $t' \leftarrow t_k \upharpoonright_{V \setminus \text{sl}_{k+1}}$ $\triangleright$ Law 3
 - 3: $t_{k+1} \leftarrow \alpha_k C_k^T t' + (1 - \alpha_k) s$ $\triangleright$ Eq. (1), Def. 8
 - 4: $\text{bal}_{k+1} \leftarrow \text{bal}_k + \text{issue}(W_k, t_k) - \text{burn}(\text{fees}_k)$
 - 5: $V_{k+1} \leftarrow \text{vset}(\Sigma_{k+1})$ $\triangleright$ Obligation 7
 - 6: return $\langle \text{rt}(t_{k+1}), \text{rt}(V_{k+1}), \text{rt}(\text{sl}_{k+1}), \text{rt}(\text{bal}_{k+1}), \text{ver} \rangle$
-

$\text{cost}(F) = O(\text{nnz}(W_k))$ adds/mults over committed data.

Line 3 is why the fixpoint is unrolled: one sparse matrix–vector product arithmetizes trivially; an eigenvector solve does not.

Definition 14 (Folding). Nova-family IVC [12]:

$$\begin{aligned} \pi_k &\vdash \Sigma_0 \xrightarrow{F^k} \Sigma_k, \\ |\pi_k| &= O(1), \quad \text{Verify}(\pi_k) = O(1), \\ \text{sync} &= \text{Verify}(\pi_k) \wedge \text{QC}_{2/3}(\Sigma_k), \\ \text{prove} &\parallel \text{consensus} \quad (\text{asynchronous, trails head}). \end{aligned}$$

Proposition 9 (Linearity).

$$\text{fin}(\Sigma_k) \text{ instant (BFT)} \Rightarrow \text{chain} \cong (\Sigma_0, \Sigma_1, \dots) \text{ linear,}$$

hence IVC applies with no re-folding branch; probabilistic-finality chains admit no such reduction.

Proof. Quorum intersection: two finalized blocks at one height require $\geq \frac{1}{3}$ of the finality committee to equivocate (Definition 12) — contradicting Definition 9 in the authority phase, and Proposition 8 with Assumption 2 in the staked phase. $\square$

Obligation 3 (Circuit discipline).

$$\begin{aligned} \forall \text{ extension } F' : \\ F' \text{ pure} \quad \wedge \quad F' \in \text{CheapArith}(\text{committed}). \end{aligned}$$

The ledger is an admission ledger, not a contract platform.

Definition 15 (Fixed-point carrier). Circuits have no reals. With precision ϕ :

$$\begin{aligned} \hat{t} &= \lfloor t \cdot 2^\phi \rfloor \in \mathbb{Z}_{\geq 0}, \\ \hat{x} \cdot \hat{y} &\mapsto \lfloor \hat{x} \hat{y} / 2^\phi \rfloor, \\ x/y &\mapsto (q, r) : x = qy + r, \\ &\quad 0 \leq r < y \quad (\text{witnessed, ranged}). \end{aligned}$$

Proposition 10 (Rounding soundness). Floor rounding is one-sided ($\hat{t}_k \leq t_k 2^\phi$: weight is never overcounted, so safety bounds survive verbatim), and the accumulated error is damped by the contraction of Proposition 4:

$$e_{k+1} \leq \bar{\alpha} e_k + n 2^{-\phi} \Rightarrow \sup_k e_k \leq \frac{n 2^{-\phi}}{1 - \bar{\alpha}},$$

bounded for all time; precision is a parameter, not a drift.

Proof. Each of the n components loses at most $2^{-\phi}$ per epoch to truncation; the prior error enters only through $\bar{\alpha} C^T$, which is ℓ_1 -nonexpansive scaled by $\bar{\alpha}$; geometric sum. $\square$

Obligation 4 (Upgrade versioning).

$$\begin{aligned} F \mapsto F' &\Rightarrow \text{vk} \mapsto \text{vk}', \\ \text{ver} &\in \Sigma, \\ \text{bind}(\text{vk}, \text{vk}') &\text{ at committed epoch;} \end{aligned}$$

shipped at genesis, not retrofitted.

Folding provides validity only: $\text{canonicity} \in \text{consensus}$, $\text{availability} \in \text{overlay}$.

Definition 16 (Commitment granularity).

$$\begin{aligned} \Gamma &\in \{\text{full}, \text{agg}, \text{blind}\}, \\ \text{leak}(\text{full}) &= \{(b, a, W[b, a])\} \quad (\text{the traffic graph}), \\ \text{leak}(\text{agg}) &= \{(v, \text{deg}^\pm(v))\}, \\ \text{leak}(\text{blind}) &= \emptyset, \quad F \text{ consumes ZK openings,} \\ \text{replay}(\Gamma) &= \top \iff \Gamma = \text{full}. \end{aligned}$$

Γ trades public replayability against surveillance of the service graph; the choice is a standing design decision, and $\Gamma = \text{blind}$ prices its privacy in circuit cost (Obligation 3).

VIII. State Relation

The ledger state machine in TLA form [13], the model-checking surface for finite instances [14]:

$$\begin{aligned} \text{Spec} &\triangleq \text{Init} \wedge \Box[\text{Next}]_\Sigma \wedge \text{WF}_\Sigma(\text{Next}), \\ \text{Init} &\triangleq \Sigma = \langle \text{rt}(s), \text{rt}(S), \emptyset, \text{rt}(\mathbf{0}), \text{ver}_0 \rangle, \\ \text{Next} &\triangleq \exists W : \text{QC}_{2/3}(\Sigma') \wedge \Sigma' = F(\Sigma, \text{rt}(W)). \end{aligned}$$

Invariant 2 (Slash monotone).

$$\text{sl} \subseteq \text{sl}', \quad v \in \text{sl} \Rightarrow t(v) = 0.$$

Proof. Lines 1–2 of Algorithm 2: sl only grows by union; the restriction $t \upharpoonright_{V \setminus \text{sl}}$ zeroes slashed carriers before every update, and Equation (1) grants no mass to $v \notin \text{carrier}$. $\square$

Invariant 3 (Version monotone).

$$\text{ver}' \geq \text{ver},$$

by Obligation 4: vk rebinding occurs only at committed epoch boundaries, forward.

Invariant 4 (Unique chain).

$$\text{fin}(\Sigma_k) \wedge \text{fin}(\Sigma'_k) \Rightarrow \Sigma_k = \Sigma'_k,$$

by Proposition 9 (quorum intersection under Assumption 2).

Proposition 11 (Progress). Under partial synchrony (after global stabilization time) with a live honest quorum — $\theta|S|$ members in the authority phase (Definition 9), honest committee majority via Proposition 8 in the staked phase —

$$\Box \Diamond \text{Next-enabled} \wedge \text{WF}_\Sigma(\text{Next}) \Rightarrow \forall k : \Diamond \text{fin}(\Sigma_k).$$

Proof. W may be empty, so Next is enabled whenever a quorum is live; BFT view-change liveness [9] delivers $\text{QC}_{2/3}$ after stabilization; weak fairness converts enabledness into occurrence. $\square$

Obligation 5 (Checkable instances). Every implementation change to F ships with a finite-instance model ($|V| \leq n_0$, bounded epochs) checking Invariants 1, 2–4 before merge.

IX. Economics

Definition 17 (Two assets).

$$\begin{aligned} \text{rank} &: \text{soulbound (Def. 7),} \\ &\quad \text{gates admission/sortition,} \\ \text{reward} &: \text{transferable,} \\ \text{issue}_k(v) &\propto (W_k^T t_k)(v). \end{aligned}$$

Proposition 12 (Countersigning game). For a service-consuming node b :

$$\begin{aligned} \text{cost}(\text{sign}) &= 0 \quad (\text{outflow spends } b\text{'s row,} \\ &\quad \text{never } t(b)), \\ u_b(\text{withhold}) &= \text{choked: service needs receipts,} \\ u_b(\text{sign}) &= \text{continued service,} \end{aligned}$$

so truthful countersigning is weakly dominant under link-layer tit-for-tat (the eDonkey/BitTorrent heritage [15], [16]). Selling receipts is not free-riding but attack-edge formation: the seller's flow enters g of Proposition 2, is volume-capped by $w_{\max}$, and is priced by Proposition 3.

Proposition 13 (Payable rank defeats cost anchoring). Let rank be transferable at price p . Then

$$\begin{aligned} \text{cost}_{\text{Adv}}(m) &= \min(\text{farm}(m), p \cdot m), \\ u_{\text{exit}}(\text{hold}) = 0 \wedge p > 0 &\Rightarrow \text{exiting nodes sell,} \\ \text{supply} > 0 &\Rightarrow p \leq \inf_v \text{farm}_v(m)/m, \end{aligned}$$

so acquisition cost decouples from behavior. Non-transferability removes the second branch of the min, restoring $\text{farm}(m) = \text{service} + \text{time} + \text{slash risk}$ as the sole path. $\square$

Invariant 5 (Balance conservation).

$$\mathbf{1}^T \text{bal}_{k+1} = \mathbf{1}^T \text{bal}_k + \text{issue}_k - \text{burn}_k,$$

by line 4 of Algorithm 2: no other term touches bal ; supply changes only through published issuance and burn.

Definition 18 (Fee loop).

$$\begin{aligned} \text{Sinks} &= \{\text{store}_{>\rho}, \text{QoS}, \text{namespace}\}, \\ \text{fees} &\rightarrow \text{burn} \cup \text{issue pool}, \\ \text{anchor}_{\text{PoS}} &\leq f(\text{demand}(\text{Sinks})). \end{aligned}$$

Obligation 6 (Security-budget crossover).

$$\text{activate}(\text{PoS}) \iff \text{Bond} \geq \beta \cdot \widehat{\text{payoff}}(\text{attack}), \quad \beta > 1,$$

conditioned, never scheduled; pre-crossover, PoA finality overlays PoS production, weighted down by $D(\text{vol}_k)$ (Definition 8).

Obligation 7 (Validator-set abstraction).

$$\begin{aligned} \text{vset} &: \Sigma \rightarrow 2^V, \\ \text{vset}_{\text{PoA}} &= \text{const } S, \\ \text{vset}_{\text{PoS}} &= \text{sortition}(\text{Bond}); \end{aligned}$$

the transition changes one pure function, not the engine.

Constraint 1 (Rejected anchors).

$$\begin{aligned} \text{PoW} &: \text{cost}(51\%) = \text{rent}(\text{hash}) \cdot O(\text{hours}) \Rightarrow \perp, \\ \text{PoT} &: W_{\text{synth}}[b, a], \quad a, b \in \text{Adv} \not\subseteq W_{\text{real}} \\ &\Rightarrow \text{bandwidth burned, } \Delta T(\text{Adv}) \approx 0 \Rightarrow \perp. \end{aligned}$$

Receipted real service with trust weighting is proof-of-traffic minus the synthetic load.

X. Security Analysis

Table I
Threat matrix

Attack	Bound	Residual
Sybil flood	Prop. 2	$\alpha g / (1 - \alpha)$
Self-dealing	Prop. 1 + signer-trust weight; cap (Def. 4)	zero-trust circulation
Patient farming	Laws 1–3; total slash (Def. 12)	undetected deviation
Whitewashing	fresh DID $\Rightarrow t = 0$; slow earn, instant loss	re-entry at zero
Grind/eclipse	Prop. 7	—
Censorship	Obl. 2	latency Δ
Seed compromise	threshold S ; revocation; Def. 8	pre-dilution window
Rank market	Prop. 13	out-of-band key sale

$$\begin{aligned} \text{sale}(\text{sk}_v) &\Rightarrow \text{buyer inherits slash exposure,} \\ \text{PoA} &= \text{trust root until Obl. 6.} \end{aligned}$$

Table II
Protocol parameters

Sym.	Role	Constraint
α_k	damping / seed weight	$\bar{\alpha} = \sup_k \alpha_k < 1$; Def. 8
$w_{\max}$	receipt issuance cap	bounds per-identity sale volume
τ	expected committee size	error $e^{-\tau D(1/3 \ T(A))}$, Prop. 8
θ	authority threshold	$\geq 2/3$; Defs. 9, 10
Δ	inclusion timeout	censorship latency, Obl. 2
μ	service metering	published per kind, Def. 4
ϕ	fixed-point precision	error $\leq n 2^{-\phi} / (1 - \bar{\alpha})$, Prop. 10
D	dilution curve	non-increasing, published, Def. 8
β	crossover margin	> 1 , Obl. 6

XI. Related Work

Local scoring descends from eDonkey queues [15] and BitTorrent choking [16]; propagation from EigenTrust and personalized PageRank [4], [5], here unrolled along epochs over signed evidence; the attack-edge bound pattern from SybilGuard/SybilLimit/SybilRank [17]–[19], instantiated over a service graph. Brahms [20] addresses adversarial sampling that the draft's tests could not (Proposition 5); DRanking sidesteps via Algorand-style sortition [6], [7] weighted by trust. S/Kademlia [21] prices identifier creation (complementary one-time cost). Finality follows PBFT lineage [9], [10]; succinctness follows Mina [11] via Nova folding [12]; impossibility framing is Douceur's [1]; the external-cost anchor observation is Nakamoto's [22].

XII. Conclusion

$$\text{safe} \iff \text{cost}(T(\text{Adv}) \geq \frac{1}{3}) > \widehat{\text{payoff}}(\text{attack}),$$

and every mechanism above — receipts, the trust laws, sortition, the ledger, folding, the two-asset economy — is a term on the left-hand side.

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